



STROKE CENTER STANDARDS

Effective: February 15, 2025

Replaces: January 1, 2021

I. Purpose

The purpose of this policy is to establish the procedures and standards for the Santa Clara County Stroke Care System. This system links pre-hospital and hospital care to ensure acute stroke patients are taken to the closest, most appropriate hospital and receive timely, optimal treatment based on these standards.

II. Authority:

California Health and Safety Code, Division 2.5, Sections 1797.114, 1797.220, 1798
California Code of Regulations Title 22, Division 9, Chapter 6.3, Article 1-5
California Code of Regulations Title 22, Division 9, Chapter 10

III. Stroke Receiving Center Designation Levels:

A. General Stroke Center:

1. Primary Stroke Center (PSC) means a hospital that treats acute stroke patients, and identified patients who may benefit from transfer to a higher level of care when clinically warranted.

B. Advanced Stroke Center

1. Thrombectomy-Capable Stroke Center (TSC) means a primary stroke center with the ability to perform mechanical thrombectomy for the ischemic stroke patient when clinically warranted.

2. Comprehensive Stroke Center (CSC) means a hospital with specific abilities to receive, diagnose and treat all stroke cases and provide the highest level of care for stroke patients.

IV. The Santa Clara County EMS Agency (EMS Agency)

A. The EMS Agency and EMS Medical Director are responsible for the oversight and management of the Stroke System including but not limited to planning, implementation and evaluation. Areas of responsibility include:

1. Designation and monitoring for compliance with standards.

2. Medical control, clinical operations, training, and management of the stroke system in the pre-hospital care setting.



3. Developing pre-hospital screening, triage and destination policies.
4. Coordinating stroke system quality improvement.

V. Designation

- A. The process for initial designation as a Stroke Center by Santa Clara County Emergency Medical Services are detailed in Policy #400.
- B. Continued designation as an EMS Stroke Receiving Center (SRC) shall depend on:
 1. Current written agreement with the EMS Agency for designation as a Stroke Receiving Center.
 2. Currently serving in the EMS System as a Pre-hospital Receiving Center (PRC).
 3. Certification from an EMS approved accreditation organization, The Joint Commission (TJC) or Det Norske Veritas (DNV).
 - a. Certification must match the level of stroke center designation.
 - b. A copy of the renewal certification shall be provided to the EMS Agency within thirty (30) days of receipt.
 4. Compliance with all State regulations and Santa Clara County EMS Agency policies and procedures including those detailed herein.
 5. If there is any conflict between National verification standards (TJC or DNV), State regulations and statutes, or local designation standards, the more stringent rule will apply until next policy review.

VI. General and Advanced Stroke Receiving Center Standards (SRC)

A designated SRC shall meet all the following standards:

- A. Hospital Facility and Services
 1. Agree to accept all EMS suspected acute stroke patients according to applicable EMS policies/protocols.
 2. Have established protocols for the triage, diagnosis, activation, and treatment of both EMS and self-transport stroke alert patients.



3. Maintain adequate number of competently trained medical staff, clinical supervisors, support staff and paraprofessionals to support stroke services offered.
4. Have a process in place to triage and treat simultaneously arriving stroke patients.
5. Maintain uninterrupted stroke capabilities which include advanced imaging, laboratory services, and treatment options in line with the requirements of their designated level twenty-four (24) hours per day, seven (7) days per week, three hundred and sixty-five (365) days per year.
6. The hospital shall maintain a single call Stroke team that may be requested by the EMS Agency for audit.
 - a. Advanced Centers must also maintain a neuro-interventional team call roster.
7. Not exceed more than 100 hours of bypass per quarter.
8. Have available within two (2) hours, neurosurgical services including operating room availability, either directly or under a written agreement with a thrombectomy-capable, or comprehensive stroke center.
9. Provide acute care rehabilitation services on-site and ensure comprehensive rehabilitation services continue either on-site or by written transfer agreement with another health care facility licensed to provide such services.

B. Staffing Requirements

1. Stroke Program Medical Director

Designate a board-certified physician in neurology, neurosurgery or another board with sufficient experience and expertise managing cerebrovascular disease. They provide medical oversight of the stroke program while also ensuring compliance with all applicable statutes, regulations and policies. They collaborate with the program coordinator for quality improvement initiatives, patient safety, medical staff training and education.

2. Stroke Program Manager/Coordinator

Designate a licensed registered nurse experienced in the clinical care of stroke patients and provides administrative support, coordination, and collaboration with the stroke program medical director to ensure quality improvement, education and training of the stroke team,



ancillary staff and patients. They are responsible for compliance with State statute, regulations, and County policy along with the program Medical Director.

3. Clinical Stroke Team

Designate a multidisciplinary group of personnel to collectively provide care to patients experiencing a stroke. The team's disciplines are determined by the designation level and at a minimum include a physician and nurse with training and expertise in acute stroke care.

4. Registrar/Data Analyst

Designate data registry personnel or analyst to enter cases into the registry, pull reports and assist with quality improvement and performance indicator metrics. It is recommended to have one full-time equivalent (1 FTE) for each 500-750 patients abstracted into the registry.

C. Stroke System Participation

1. Participate in the pre-hospital performance improvement process and offer expertise to the EMS Medical Director in the development of pre-hospital care protocols relevant to stroke patients.
2. Participate in the Stroke Care System Quality Improvement Committee (SCSQIC) as per Policy #441 and other related committees when requested.
3. Participate in the pre-hospital quality improvement process through selected review of stroke-related deaths, major complications, and transfers when requested by EMS Agency.
 - a. The Stroke Program Manager or designee from each designated facility will work with the EMS Agency Specialty Programs Nurse Coordinator to obtain the necessary chart materials for the medical review process and committee meetings.
4. For any Stroke Center receiving a Stroke patient as a result of ED Bypass or Stroke Bypass shall conduct a quality review on the patient and report any adverse finding to the EMS Agency.

D. Data Collection and Submission.



1. The EMS Agency will be responsible for creating a *Santa Clara County Stroke Data Dictionary* to include the required local, state, and national data elements and will be updated annually.
2. Stroke Receiving Center shall:
 - a. Collect the minimum data elements identified in the *Santa Clara County Stroke Data Dictionary* directly into an audited registry approved by the EMS Agency.
 - b. Submit the data in an agreed-upon format to the EMS Agency on regularly assigned dates.
 - c. Ensure that data is validated and without error prior to submission.
3. Stroke data will be submitted to the California EMS Information System as required by State regulations.

E. Performance Standards

A designated Stroke Receiving Center shall meet all the following performance standards:

1. Have a process to review and revise facility stroke policies and protocols at least every three (3) years.
2. Have a data-driven quality improvement program that monitors standardize clinical performance measures which align with the American Heart Association and certification requirements.
3. Additional performance measures as determined by the EMS Agency and/or the SCSQIC.

F. Education

1. Provide pre-hospital stroke continuing education opportunities at least once per year for EMS personnel and report these activities to the EMS Agency by the end of each calendar year.
2. Ensure staff physicians, staff nurses, staff allied health personnel, and members of the stroke clinical team receive continuing education as required by certification level.



3. Have a patient population focused community outreach and education campaign that details about stroke warning signs and the importance of early utilization of the 9-1-1 system. Activity must be reported by the end of each calendar year to the EMS Agency.
4. Advanced Stroke Centers shall provide stroke-specific medical education and guidance to PSC with which they maintain transfers agreements.
5. CSC stroke centers shall have a stroke patient research program.

G. Reporting Requirements

1. Shall notify the EMS Agency Duty Chief as soon as possible in the case of an unplanned event causing the CT scanner or neurointerventional to be unavailable as well as when the issue is resolved.
2. Shall notify the EMS Agency through the submission of a System Variance Report (SVR) at least 24 hours prior to any planned event that will result in the CT scanner or neurointerventional services being unavailable.
3. Shall notify the EMS Agency in writing of any failure to meet these, *EMS Stroke Center Standards* within 48 hours.
4. Changes to key SRC personnel shall be reported to the EMS Agency within 10 (ten) business days to include:
 - a. Stroke Medical Director
 - b. Stroke Program Manager

VII. References

- A. DNV. (n.d.). *Stroke care certification programs*. Retrieved January 21, 2025, from <https://www.dnv.us/services/stroke-care-certification-programs-219582/>
- B. The Joint Commission. (2021). *Certification by setting: Stroke comparison chart*. Retrieved from <https://www.jointcommission.org/-/media/tjc/documents/accred-and-cert/certification/certification-by-setting/stroke/dsc-stroke-grid-comparison-chart-42021.pdf>
- C. The Joint Commission. (n.d.). *Disease-specific care certification: Review process guide*. Retrieved January 21, 2025, from <https://www.jointcommission.org/-/media/tjc/documents/review-process-guide/disease-specific-care-organization-rpg.pdf>